

María González Rodríguez

Aerospace Engineer & Industrial Mathematics MSc Student | Simulation Intern @ European Space Agency - Spaceship Poland
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Summary

Aerospace Engineer and Industrial Mathematics Master's student with experience in spacecraft-related modelling, mission-planning optimisation, AOCS sizing and numerical simulation, using MATLAB and Python. Previous work includes rotational-motion analysis for microgravity simulation, mission-planning optimisation for Near-Earth asteroid flybys, AOCS sizing in ESA's Concurrent Engineering Workshop and parameterised lunar terrain generation. I am interested in applying mathematical modelling and numerical analysis to spacecraft dynamics, trajectory planning and control-related analysis, especially where model uncertainty affects mission design and safety.

Education

Master's Degree in Industrial Mathematics, Polytechnic University of Madrid September 2024 – Present

- Proficient in MATLAB and COMSOL Multiphysics for modelling and simulation.
- **Key Courses:** Ordinary Differential Equations & Dynamical Systems; Stability of Physical Systems; Reduced-Order Modelling; Numerical Methods & Programming (MATLAB); Multidisciplinary Design Optimization.

Bachelor's Degree in Aerospace Engineering, Polytechnic University of Madrid September 2018 – September 2023

- **Key Courses:** Flight Mechanics; Orbital Mechanics; Automatic Control; Aircraft Dynamics & Stability; Numerical Methods.
- **Bachelor's thesis:** *ORBITMiNER: Optimal Routing Biological-Inspired Techniques for Mining of Near-Earth Resources*, awarded the highest distinction.

Technical Skills

- **Programming:** MATLAB, Simulink, Python, Fortran.
- **Simulation and Modelling Tools:** COMSOL Multiphysics, Unreal Engine 5.
- **Engineering Tools:** CDP4-COMET, CATIA V5, LaTeX.

Projects

Random Positioning Machine for Microgravity and Partial-Gravity Simulation 2022

- **Goal:** Simulate microgravity and partial-gravity environments through controlled rotational motion.
- **Contribution:** Designed and analysed rotational algorithms, evaluating residual acceleration and simulated gravity-vector behaviour to assess the quality of the generated gravity conditions.

ORBITMiNER: Optimisation-Based Mission Planning for Near-Earth Asteroid Flybys 2023

- **Goal:** Model asteroid flyby sequencing as a time-dependent mission-planning problem with moving Near-Earth asteroid targets.
- **Contribution:** Developed and tested genetic algorithm strategies in Python, evaluating performance across heuristics, constraints and mission-design assumptions.

Concurrent Engineering Workshop, European Space Agency 2025

- **Goal:** Develop a preliminary spacecraft concept for a solar sail Jupiter flyby mission in a multidisciplinary concurrent engineering environment.
- **Contribution:** Performed AOCS sizing and constraint analysis, translating mission-level requirements into attitude-control-relevant subsystem constraints and preliminary design trades.

Lunar Solar Panel Placement: Multidisciplinary Simulation and Optimisation 2025

- **Goal:** Evaluate lunar solar panel placement by modelling environmental effects on power generation and thermal behaviour.
- **Contribution:** Developed and validated a multidisciplinary simulator, then used numerical experiments, correlation matrices, sensitivity analysis and gradient-based, heuristic and multi-objective optimisation methods to study design trade-offs.

Experience

Research Intern, ESA Spaceship Poland – Poznań, Poland

February 2026 – Present

- Developed the science-parameterised terrain generation module for an Unreal Engine 5 lunar environment simulator within a ROS 2-integrated rover simulation framework.
- Modelled crater populations, degradation states, ejecta zones and rock placement from statistical and geological descriptions of the lunar surface.
- Produced co-registered terrain analysis layers, including elevation, slope, crater/ejecta classifications and spatial density maps, and validated outputs against lunar surface literature.

ATM Systems Engineer, ISDEFE – Madrid, Spain

March 2025 – January 2026

Supported requirements review, technical documentation and verification/validation activities for safety-critical ATM systems.

ATM-CNS Engineer, INECO – Madrid, Spain

March 2024 – March 2025

Performed technical studies and simulations to assess CNS infrastructure performance against operational and engineering requirements.

Intern, Asteroid Mining Corporation Ltd – London, United Kingdom

July 2022 – October 2022

- Supported early-stage mission concept development for a Near-Earth asteroid exploration probe.
- Updated spacecraft-level budget estimates for mass, power and cost to support feasibility assessment and design trade-offs.
- Prepared technical documentation summarising mission assumptions, design choices and preliminary feasibility results.

Publications

Scientifically Grounded Lunar Terrain Generation for High-Fidelity Rover Perception and Navigation Simulation

Submitted to i-SAIRAS/iSpaRo 2026 | Co-author

Contributed the terrain generation module, including statistically grounded crater, ejecta and rock-placement modelling, co-registered analysis layers and validation against lunar surface literature.

Development of an Improved Random Positioning Machine to Simulate Organic Growth in Microgravity

73rd International Astronautical Congress | Co-author

Co-authored work on rotational-pattern design for microgravity simulation, focused on reducing residual acceleration and improving the fidelity of simulated space environments.

Literature Studies and Experimental Characterization of Multiple Solid Propellant Regression Rates Using Crawford Bomb Method

71st International Astronautical Congress | Co-author

Contributed to the experimental characterization of solid propellant mixtures through Crawford bomb method measurements.

Languages

- **Spanish:** Native
- **English:** C1